

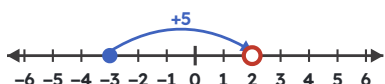
Step 1 Adding a positive Easy

HOW TO

Start at the first number.

Adding a positive number: move right.

$-3 + 5$: start at -3 , move 5 right.
Land on 2.



Draw the jump. Write where you land.

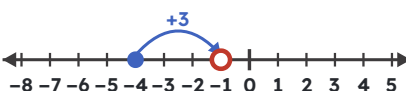
WE DO

E1 $-2 + 4$



YOU DO

1 $-4 + 3$



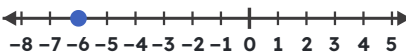
$= -1$

2 $-5 + 2$



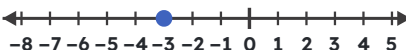
$=$ ____

3 $-6 + 4$

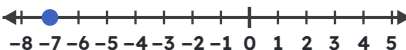


$=$ ____

4 $-3 + 7$



5 $-7 + 7$



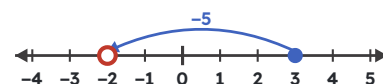
Step 2 Adding a negative Easy

HOW TO

Adding a negative number: move left.

$+(-4)$ means the same as -4 .

$3 + (-5) = 3 - 5 = -2$



Rewrite $+(-)$ as $-$. Then work it out.

WE DO

E2 $6 + (-2)$

E3 $-2 + (-3)$

YOU DO

6 $7 + (-3)$

7 $5 + (-5)$

$= 7 - 3$

$= 5 - 5$

$= 4$

$=$ ____

8 $2 + (-6)$

9 $4 + (-10)$

$=$ ____ $-$ ____

$=$ ____

10 $-1 + (-4)$

11 $-3 + (-6)$

Step 3 Zero pairs Medium

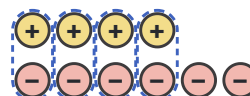
HOW TO

A $+$ and a $-$ make a zero pair.

They cancel out.

So take the difference. The sign with more wins.

$-6 + 4$: $6 - 4 = 2$. $-$ has more: -2 .



Take the difference. Which sign has more?

WE DO

E4 $-5 + 8$

E5 $6 + (-9)$

YOU DO

12 $-7 + 10$

13 $-9 + 4$

$10 - 7 = 3$

$9 - 4 = 5$

$+$ has more: 3

$-$ has more: ____

14 $8 + (-12)$

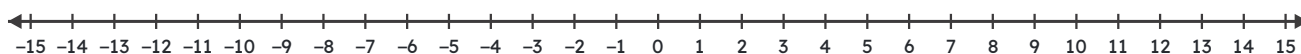
15 $-15 + 20$

____ $-$ ____ $=$ ____

$-$ has more: ____

16 $25 + (-40)$

17 $-36 + 50$



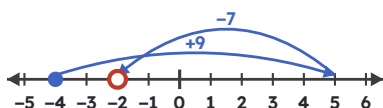
Step 4 Three numbers

Medium

HOW TO

Add from left to right, one jump at a time.

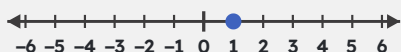
$$-4 + 9 + (-7) = 5 + (-7) = -2$$



Add from left to right. Draw the jumps when there is a number line.

WE DO

E6 $1 + (-4) + 5$



YOU DO

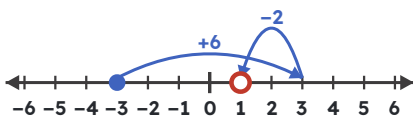
18 $2 + (-5) + 4$



$$= -3 + 4$$

$$= 1$$

19 $-3 + 6 + (-2)$



$$= 3 + (-2)$$

$$= 1$$

20 $12 + (-15) + (-6)$

$$= 12 + (-21)$$

$$= -9$$

21 $-20 + 8 + 15$

Step 5 Adding in real life

Challenging

HOW TO

Write the story as a sum.

Up, rises and gains are **positive**.

Down, falls and spending are **negative**.

-6°C , then it warms up 10°C :

$$-6 + 10 = 4^{\circ}\text{C}$$

Write a sum. Then answer the question.

WE DO

E7 It is -5°C . The temperature rises 8°C . What is the temperature now?

YOU DO

22 It is -4°C at dawn. By lunch it has warmed up 9°C . What is the temperature at lunch?

$$-4 + 9$$

$$= 5^{\circ}\text{C}$$

23 A lift is at level -2 . It goes up 5 floors. Which level is it on now?

$$-2 + 5$$

$$= \text{level } 3$$

24 Sam has \$15 in the bank. Sam spends \$40 using credit. What is the balance now?

$$15 + (-40)$$

$$= -25$$

25 A diver is at -18 m . She swims up 7 m , then down 4 m . Where is she now?

Step 6 Is it right?

Challenging

HOW TO

Work it out yourself first.

Then compare. Say yes or no, and give the right answer.

Kim says $3 + (-5) = -2$.

$3 - 5 = -2$. Yes, Kim is right.

Is it right? Explain your answer.

WE DO

E8 Tom says $-6 + 2 = -8$. Is he right?

YOU DO

26 Mia says $-8 + 5 = -13$. Is she right?

start at -8 , move 5 right

$$= -3$$

No. The answer is -3 .

27 Ben says $-7 + 3 + (-3) = -7$. Is he right?

$3 + (-3)$ is a zero pair

$$-7 + 0 = -7$$

28 Ali says $-4 + \square = 2$ means $\square = -2$. Is he right?

from -4 to 2 is 6 right

$$\square = 6$$

No. $\square$ is 6 , not -2 .

29 Jo says adding always makes a number bigger. Is Jo right?